

Final Example
Times: 90 minutes

Question 1 (2 points). Let $\phi = \exists z(P(x, z) \wedge (\forall y (\neg Q(y, x) \vee P(y, t))))$ and $\psi = \forall z P(y, z) \rightarrow (\forall x \forall y (\neg Q(y, x) \vee R(y, z)))$.

- a) Draw parse trees and identify the free and bound variables for ϕ and ψ .
- b) Determine $\phi[(f(y)/x)]$ and $\psi[z/y]$.

Question 2 (2 points). Verify the following sequents using natural deduction rules.

- a) $\exists x(P(x) \wedge Q(x)), \forall x(Q(x) \rightarrow R(x)) \vdash \exists x(P(x) \wedge R(x))$
- b) $\exists x P(x), \forall x \forall y (P(x) \rightarrow Q(y)) \vdash \forall y Q(y)$

Question 3 (2 points). For each of the formulas below, find a model that satisfies the formula and one that does not.

- a) $\forall x(P(x) \rightarrow Q(f(x)))$
- b) $\exists x \forall y((P(x, y) \wedge P(y, x)) \vee (Q(x, y) \wedge R(y, x)))$

Question 4 (2 points). Which of the following semantic entailments hold?

- a) $\exists x P(x) \wedge \exists x Q(x) \models \exists x (P(x) \wedge Q(x))$
- b) $\exists y \forall x P(x, y) \models \forall x \exists y P(x, y)$

Question 5 (2 points). Use the tableau algorithm to determine whether the following formulas is valid or not.

- a) $\phi = ((p \rightarrow q) \wedge p) \vee q$
- b) $\psi = \forall x \neg P(x) \vee \exists x (P(x) \vee Q(x))$